

Total marks (84)**Attempt questions 1 – 7****All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Use a SEPARATE writing booklet.	Marks
a) Factorise $a^2 - 4b^2 + 2a + 4b$.		2
b) Solve $x^2 > 9x$.		2
c) Solve the following inequation		
$\frac{x+1}{x-3} \leq 2.$		3
d) Using the fact that $(\sqrt{x} + \sqrt{y})^2 = x + y + 2\sqrt{xy}$, express $11 + 4\sqrt{6}$ in the form $(\sqrt{x} + \sqrt{y})^2$.		2
e) Find the values of a, b, c given that $x^2 + x + 1 \equiv a(x-1)^2 + b(x-1) + c$.		3
Question 2 (12 marks)		
Use a SEPARATE writing booklet.		
a) By drawing appropriate straight lines on the same axes as the graph of $y = 2^x$, determine the number of solutions for each of the following equations:		3
(i) $2^x = x + 2$		
(ii) $2^x = x - 2$		
(iii) $2^x = 2 - x$		
b) (i) Sketch the graphs of $y = x^2 - 2 $ and $y = 3$ on the same number plane.		2
(ii) Write down the inequalities that simultaneously represent the region bounded by the two graphs.		2
c) A parabola has the equation $x^2 - 10x + 16y + 9 = 0$.		
(i) Express this equation in the form $(x-h)^2 = -4a(y-k)$.		2
(ii) Draw a neat sketch of the parabola indicating the coordinates of the vertex and focus.		3

Question 3 (12 marks) Use a SEPARATE writing booklet.**Marks**

- a) The line $3x + ky - 5 = 0$ makes an angle of 135° with the positive x axis.
- (i) Show that the gradient of the line is $-\frac{3}{k}$. **1**
- (ii) Find the value of k . **2**
- b) Find the coordinates of the point that divides the join of $(-3, 2)$ and $(5, 7)$ **externally** in the ratio $1 : 3$. **3**
- c) Find the perpendicular distance between the pair of parallel lines $3x - 4y + 12 = 0$ and $3x - 4y - 6 = 0$. **3**
[Hint: Find the coordinates of a point on one of the lines.]
- d) Prove that $x^2 + (k - 3)x - k = 0$ has real roots for all values of k . **3**

Question 4 (12 marks) Use a SEPARATE writing booklet.

- a) A point $P(x, y)$ moves in such a way that its distance from $A(-2, 1)$ is twice its distance from the point $B(2, 1)$. Determine and describe the locus of the point $P(x, y)$. **4**
- b) Consider the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$.
- (i) Show that the equation of the normal to the parabola $x^2 = 4ay$ at the point P is given by $x + py = 2ap + ap^3$. **2**
- (ii) Find the equation of the line which passes through the focus $S(0, a)$ and is perpendicular to the normal. **2**
- (iii) If the line found in part (ii) meets the normal at N , find the coordinates of N . **2**
- (iv) Show that the locus of N is a parabola and find its vertex. **2**

Question 5 (12 marks) Use a SEPARATE writing booklet.**Marks**

a) Solve $\sin 2\theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 3

b) If $\sin \theta = p$, where θ is acute, find expressions for $\cos \theta$ and $\tan \theta$ in terms of p . 3

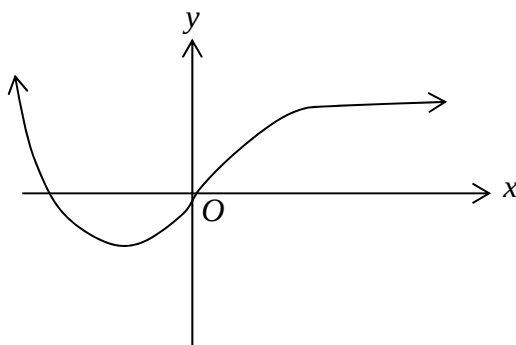
c) Solve for $0^\circ \leq \theta \leq 360^\circ$, to the nearest degree: $4\sec^2 \theta - \tan \theta = 7$. 3

d) Solve for x :

$$x^2 - 5x + 2 + \frac{4}{5x - x^2 - 2} = 0. \quad \text{3}$$

Question 6 (12 marks) Use a SEPARATE writing booklet.

a) 2



The diagram shows the graph of $y = f(x)$.

Copy or trace this graph into your Writing Booklet.

On the same set of axes draw the graph of $y = f'(x)$.

b) Let $y = x\sqrt{x}$.

(i) Find $\frac{dy}{dx}$, leaving your answer in surd form. 2

(ii) Find the coordinates of the point on the graph of $y = x\sqrt{x}$ at which the normal has a slope of $-\frac{1}{4}$. 3

Question 6 continued.**Marks**

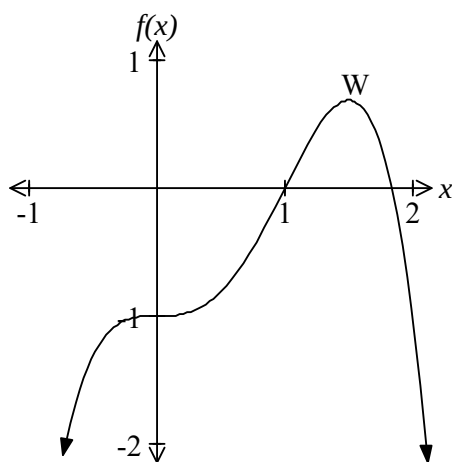
c) If $f(x) = \frac{x-1}{\sqrt{2x+1}}$,

(i) show that $f'(x) = \frac{x+2}{\sqrt{(2x+1)^3}}$. **2**

(ii) Show that $f(x)$ is an increasing function in the domain of x . **3**

Question 7 (12 marks) Use a SEPARATE writing booklet.

a) Study the graph of $y = ax^4 + bx^3 + c$ shown below.



(i) Given that $(1, 0)$ is a point of inflection, evaluate the constants a , b and c . **3**

(ii) Find the co-ordinates of W (the absolute maximum of the function). **2**

b) Given that $y = \frac{2}{(x-1)(x-3)}$,

(i) find the equations of all (horizontal and vertical) asymptotes **2**

(ii) find the y -intercept **1**

(iii) find the co-ordinates of any stationary point(s) and the nature of the stationary point(s). (You need not calculate the second derivative.) **2**

(iv) sketch the function showing all significant information. **2**

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note $\ln x = \log_e x, \quad x > 0$

Q1

Yr 11 Ext 1 2007

$$a) \quad (a-2b)(a+2b) + 2(a+2b) \checkmark$$

$$= (a+2b)(a-2b+2) \checkmark$$

$$b) \quad x^2 > 9x \quad x < 0, x > 9 \checkmark$$

$$x^2 - 9x > 0$$

$$x(x-9) > 0 \checkmark$$

$$c) \quad \frac{x+1}{x-3} \leq 2 \quad x < 3 \text{ or } x \geq 7 \checkmark$$

$$\frac{x+1}{x-3} \leq 2 \quad x(x-3)^2 \quad x(x-3)^2$$

$$(x+1)(x-3) \leq 2(x-3)^2 \checkmark$$

$$0 \leq 2(x-3)^2 - (x+1)(x-3)$$

$$0 \leq (x-3)(2x-6-x-1) \checkmark$$

$$0 \leq (x-3)(x-7) \checkmark$$

$$d) \quad 11 + 4\sqrt{6} = 11 + 2\sqrt{24} \quad x = 8 \quad y = 3 \checkmark$$

$$x + y = 11 \quad xy = 24 \checkmark$$

$$e) \quad x^2 + x + 1 = a(x-1)^2 + b(x-1) + c$$

$$\text{let } x=0 \quad 1 = a - b + c$$

$$\text{let } x=2 \quad 7 = a + b + c$$

$$\text{let } x=1 \quad 3 = c \quad \checkmark$$

$$a - b = -2 \quad (1)$$

$$a + b = 4 \quad (2)$$

$$2a = 2$$

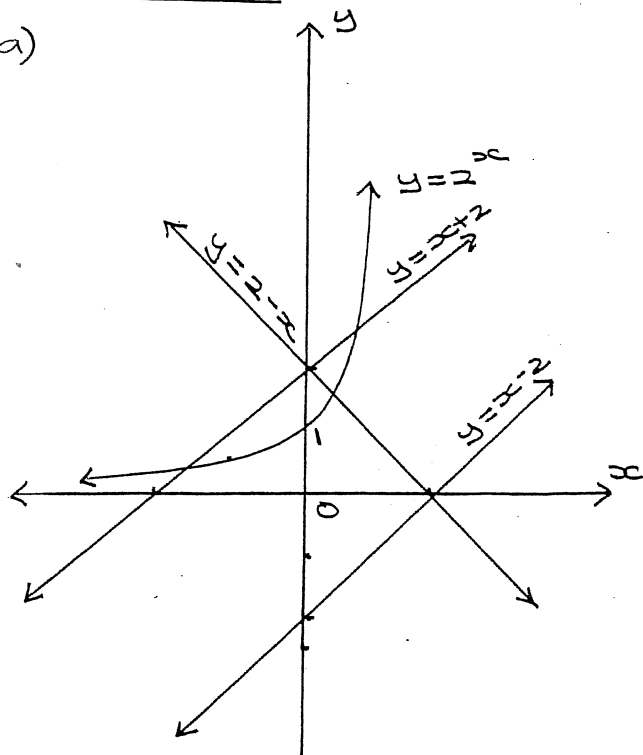
$$a = 1 \quad \checkmark$$

$$b = 3 \quad \checkmark$$

* one mark is allocated to method

Question 2

a)

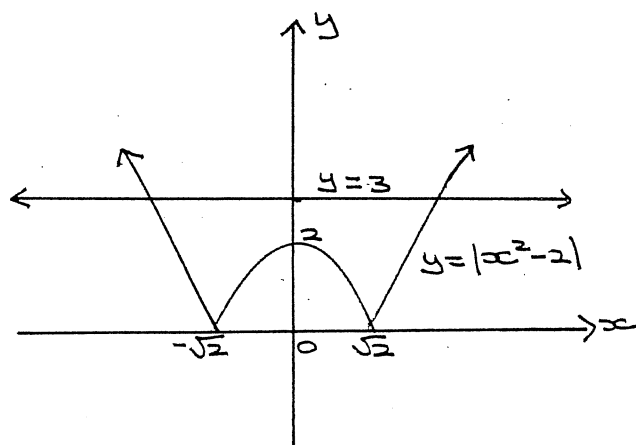


(i) 2

(ii) 0

(iii) 1

b) i)



$$y = |x^2 - 2| = 3$$

$$x^2 - 2 = 3$$

$$x^2 = 5$$

$$x = \pm \sqrt{5}$$

$$0 \leq y \leq 3 \quad -\sqrt{5} \leq x \leq \sqrt{5}$$

① Marks . . . if answers were given without diagrams

① Mark if correct diagram was given without answers

③ Marks for correct answers + diagram

① Mark for each graph.

For the graph $y = |x^2 - 2|$
 $-\sqrt{2}$ and $\sqrt{2}$ had to be shown and the y-intercept

① Mark for each inequality

c) $x^2 - 10x + 16y + 9 = 0$

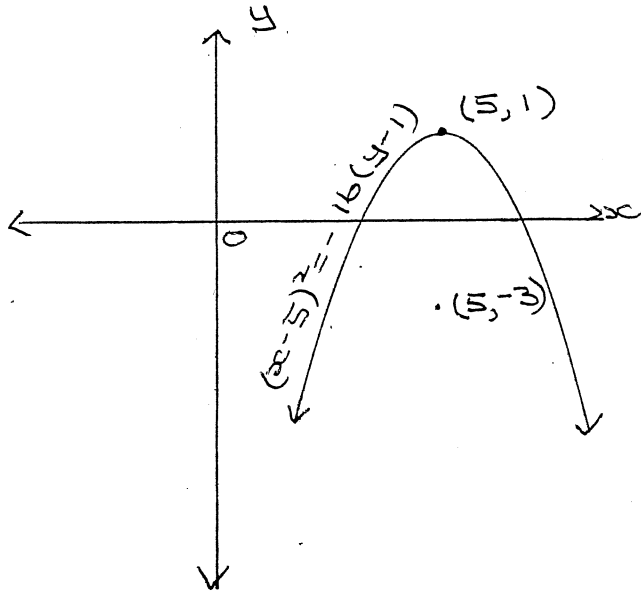
$x^2 - 10x + 25 = -16y - 9 + 25$ — ① Mark

$(x-5)^2 = -16y + 16$

$(x-5)^2 = -16(y-1)$ — ① Mark

vertex = $(5, 1)$

focal length = 4



① Mark for vertex

① Mark for $\curvearrowright$ and shape

① Mark for vertex

Question 3

(a) $3x + ky - 5 = 0$

(i) $3x + ky - 5 = 0$

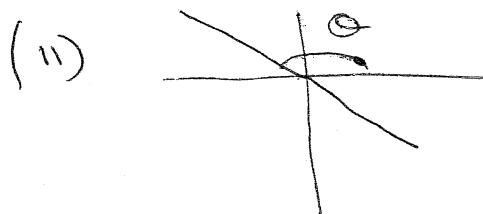
$ky = -3x + 5$

$y = -\frac{3}{k}x + \frac{5}{k}$ ✓

(1 mark)

answer in gradient-intercept form.

∴ the gradient is $-\frac{3}{k}$



$\theta = 135^\circ$

$\tan \theta = m$

$\tan 135^\circ = -\tan 45^\circ$

$= -1$ ✓

1 mark for $\tan 135^\circ = -1$

∴ $-\frac{3}{k} = -1$

∴ $\underline{\underline{k = 3}}$

✓ (2 marks) for k.

(b) $(-3, 2)$ $(5, 7)$ externally 1: $\textcircled{-3}$ ✓

1 mark for neg value.

$P(x, y) = \left(\frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right)$

1 mark correct sub.

$= \left(\frac{1 \times 5 + -3 \times -3}{1 + -3}, \frac{1 \times 7 + -3 \times 2}{1 + -3} \right)$ ✓

$= \left(\frac{5+9}{-2}, \frac{7-6}{-2} \right)$

$= (-7, -\frac{1}{2})$

1 mark correct evaluation of your values sub.

✓ (3 marks)

(c) $3x - 4y + 12 = 0$

$3x - 4y - 6 = 0$

when $y=0$ $x = -4$

∴ passes through $(-4, 0)$ ✓

1 mark Finding a point on a line

Perpendicular distance from $(-4, 0)$ to $3x - 4y - 6 = 0$

$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$= \frac{|3x - 4 + -4 \times 0 - 6|}{\sqrt{3^2 + (-4)^2}}$ ✓

1 mark for correct sub of your values.

Question 3

(c) $d = \frac{1-12-61}{5}$

$d = \frac{18}{5}$ ✓

1 mark correct evaluation of your values

3 (3 marks)

(d) $x^2 + (k-3)x - k = 0$

Real roots $\Delta \geq 0$

(3 marks)

$\therefore b^2 - 4ac \geq 0$

$(k-3)^2 - (4 \times 1 \times -k) \geq 0$ ✓

1 mark correct sub into Δ

$k^2 - 6k + 9 + 4k \geq 0$

$k^2 - 2k + 9 \geq 0$

$k^2 - 2k + 1 + 8 \geq 0$

$(k-1)^2 + 8 \geq 0$

✓ 1 mark correct formation of perfect square.

$(k-1)^2 \geq 0$ For all values of k

So $(k-1) + 8 \geq 0$ For all values of k

$\therefore x^2 + (k-3)x - k = 0$ has real roots for all values of k .

✓ 1 mark correct conclusion

Question 4

(a) A(-2, 1) B(2, 1)

Let P(x, y) be any point of the locus.

Condition of locus

$$* AP = * 2BP$$

$$\sqrt{(x+2)^2 + (y-1)^2} = 2\sqrt{(x-2)^2 + (y-1)^2}$$

$$(x+2)^2 + (y-1)^2 = 4((x-2)^2 + (y-1)^2)$$

$$x^2 + 4x + 4 + y^2 - 2y + 1 = 4(x^2 - 4x + 4 + y^2 - 2y + 1)$$

$$x^2 + 4x + y^2 - 2y + 5 = 4x^2 - 16x + 16 + 4y^2 - 8y + 4$$

$$3x^2 - 20x + 3y^2 - 6y + 15 = 0$$

$$x^2 - \frac{20}{3}x + y^2 - 2y + 5 = 0$$

$$x^2 - \frac{20}{3}x + \frac{100}{9} + y^2 - 2y + 1 = -5 + \frac{100}{9} + 1$$

$$(x - \frac{10}{3})^2 + (y-1)^2$$

$$(x - \frac{10}{3})^2 + (y-1)^2$$

$\therefore$ circle centre $(\frac{10}{3}, 1)$

radius = $\frac{8}{3}$

1 mark for description and details.

1 mark correct condition of locus.

1 mark for correct expansion + putting on LHS equal to 0

1 mark complete both squares correctly.

(b)(i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

Gradient of tangent at P($2ap$, ap^2)

$$\therefore m = p$$

Gradient of normal at P

$$\therefore m = -\frac{1}{p}$$

Equation of a normal

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$\therefore x + py = 2ap + ap^2$ is the required equation.

1 mark for finding correctly the gradient.

1 mark for correct derivation of the normal.

(4)

(2)

Question 4

(b) (ii) equation of line through $S(0, a)$ $m=p$ ✓

$$y-a = p(x-0)$$

$$\therefore y = px + a$$
 ✓

1 mark for gradient

(2)

1 mark for correct equation.

(iii) Solve simultaneously

$$y = px + a \quad \text{--- (1)}$$

$$x + py = 2ap + ap^3 \quad \text{--- (2)}$$

Sub (1) into (2)

$$x + p(px + a) = 2ap + ap^3$$

$$x + p^2x + pa = 2ap + ap^3$$

$$x(1 + p^2) = ap^3 + ap$$

$$x = \frac{ap(p^2 + 1)}{(1 + p^2)}$$

$$\therefore x = ap \quad \text{--- (3)} \quad \checkmark \quad \text{check in (2)}$$

Sub (3) in (1)

$$y = p(ap) + a$$

$$y = ap^2 + a$$

$$ap + p(ap^2 + a)$$

$$= ap + ap^3 + ap \quad \checkmark$$

1 mark for y value but ~~2x~~ + y had to be simplified.

$\therefore N$ has the coordinates $(ap, ap^2 + a)$

IV Locus of N $(ap, ap^2 + a)$

$$\therefore x = ap$$

$$p = \frac{x}{a}$$

$$a = \frac{x}{p}$$

$$y = ap^2 + a$$

$$y = a\left(\frac{x}{a}\right)^2 + a$$

$$y = \frac{x^2}{a} + a$$

1 mark for the equation of the locus.

$\therefore$ The locus is a parabola.

The vertex is $(0, a)$

1 mark for the vertex. ✓

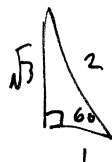
(2)

Q5

a)

$$2\theta = 60, 120, 420, 480$$

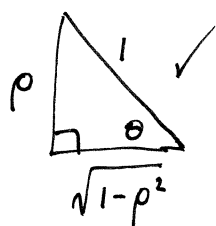
$$\therefore \theta = 30, 60, 210, 240$$



b)

$$\sin \theta = p$$

$$\cos \theta = \sqrt{1-p^2}$$



$$\tan \theta = \frac{p}{\sqrt{1-p^2}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

c)

$$4 \sec^2 \theta - \tan \theta = 7$$

$$4(\tan^2 \theta + 1) - \tan \theta - 7 = 0$$

$$4 \tan^2 \theta - \tan \theta - 3 = 0$$

$$(4 \tan \theta + 3)(\tan \theta - 1) = 0$$

$$\tan \theta = -\frac{3}{4}$$

$$\theta = 143^\circ \text{ or } 323^\circ$$

$$\tan \theta = 1$$

$$\theta = 45^\circ \text{ or } 225^\circ$$

d) Let $u = x^2 - 5x + 2$

$$u + \frac{4}{-u} = 0$$

$$-u^2 + 4 = 0$$

$$u^2 - 4 = 0$$

$$u = \pm 2$$

$$x^2 - 5x + 2 = 2$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0$$

$$x = 0 \text{ or } 5$$

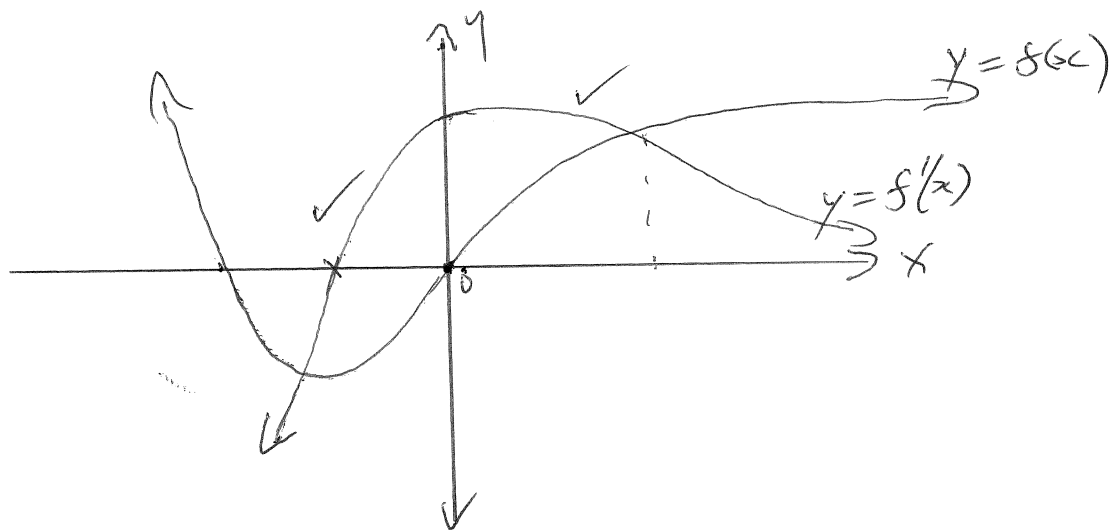
$$x^2 - 5x + 2 = -2$$

$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

$$x = 4 \text{ or } 1$$

Q6 (a)



(b) (i) $y = x\sqrt{x}$
 $= x^{3/2}$ ✓
 $\frac{dy}{dx} = \frac{3}{2}x^{1/2}$
 $= \frac{3\sqrt{x}}{2}$ ✓

(ii) If Normal has $m = -\frac{1}{4}$ ✓
 Tangent has $m = 4$

$\therefore$ let $\frac{3\sqrt{x}}{2} = 4$
 $\sqrt{x} = \frac{8}{3}$
 $x = \frac{64}{9}$ ✓

$\therefore$ Coordinates are $(\frac{64}{9}, \frac{512}{27})$ ✓
 (18.96)

(c) (i) $f(x) = \frac{x-1}{\sqrt{2x+1}}$ ✓
 $f'(x) = \frac{\sqrt{2x+1} \cdot 1 - (x-1) \cdot \frac{1}{\sqrt{2x+1}}}{(\sqrt{2x+1})^2}$ ✓
 $= \frac{(2x+1) - (x-1)}{(2x+1)\sqrt{2x+1}}$
 $= \frac{x+2}{\sqrt{(2x+1)^3}}$ ✓

(ii) Show that $f'(x) > 0$ in the domain of x

Domain $2x+1 > 0$
 $x > -\frac{1}{2}$ ✓

Now as $x+2 > 0$ if $x > -\frac{1}{2}$ ✓
 $2x+1 > 0$ if $x > -\frac{1}{2}$ ✓

$\therefore f'(x) = \frac{x+2}{\sqrt{(2x+1)^3}} > 0$

Note: Marks were awarded if students correctly attempted to find $f'(x)$ & nature.

$$7a) \quad y = ax^4 + bx^3 + c$$

$$(i) \quad \frac{dy}{dx} = 4ax^3 + 3bx^2$$

$$\frac{d^2y}{dx^2} = 12ax^2 + 6bx$$

$$\text{let } \frac{d^2y}{dx^2} = 0 \text{ for Ptof inflection}$$

$$12ax^2 + 6bx = 0$$

$$6x(2ax + b) = 0$$

$$x = -\frac{b}{2a}$$

$$\text{let } x=1 \quad \therefore 2a = -b \text{ or } b = -2a$$

$$\text{as } c = -1 \text{ [from graph]} \quad \checkmark$$

$$\therefore P(1,0) \text{ satisfies } y = ax^4 + bx^3 + c$$

$$\therefore 0 = a + b - 1$$

$$\therefore a + b = 1$$

$$\therefore a - 2a = 1$$

$$a = -1 \quad \checkmark$$

$$b = 2 \quad \checkmark$$

$$c = -1 \quad \checkmark$$

$$\left[\begin{array}{l} \text{If } (2,0) \text{ is used} \\ \text{then } c = -1, a = -\frac{7}{8}, b = \frac{15}{8} \end{array} \right.$$

$$(ii) \quad \text{let } \frac{dy}{dx} = 0$$

$$4ax^3 + 3bx^2 = 0$$

$$\therefore -4x^3 + 6x^2 = 0$$

$$-2x^2(2x - 3) = 0$$

$$\therefore x = 0, \frac{3}{2}$$

$$\therefore W\left(\frac{3}{2}, \frac{11}{16}\right) \quad \checkmark$$

$$\text{Take } x = \frac{3}{2} \quad \checkmark$$

$$\therefore y = -1\left(\frac{3}{2}\right)^4 + 2\left(\frac{3}{2}\right)^3 - 1$$

$$= -\frac{81}{16} + \frac{27}{4} - 1$$

$$7b) \quad y = \frac{2}{(x-1)(x-3)}$$

(i) Vertical asymptotes: $x=1$, $x=3$ ✓ (Both)

Horizontal asymptote: $y=0$ ✓

[2]

(ii) y -int occurs when $x=0$

$$y = \frac{2}{-1 \times -3} = \frac{2}{3}$$

✓

[1]

$$(iii) \quad \frac{dy}{dx} = \frac{(x-1)(x-3) \cdot 0 - 2[(x-1) \cdot 1 + (x-3) \cdot 1]}{(x-1)^2(x-3)^2}$$

$$= \frac{-2[2x-4]}{(x-1)^2(x-3)^2}$$

$$= \frac{-4[x-2]}{(x-1)^2(x-3)^2}$$

✓

[2]

St pts occur when $\frac{dy}{dx} = 0 \therefore x=2$

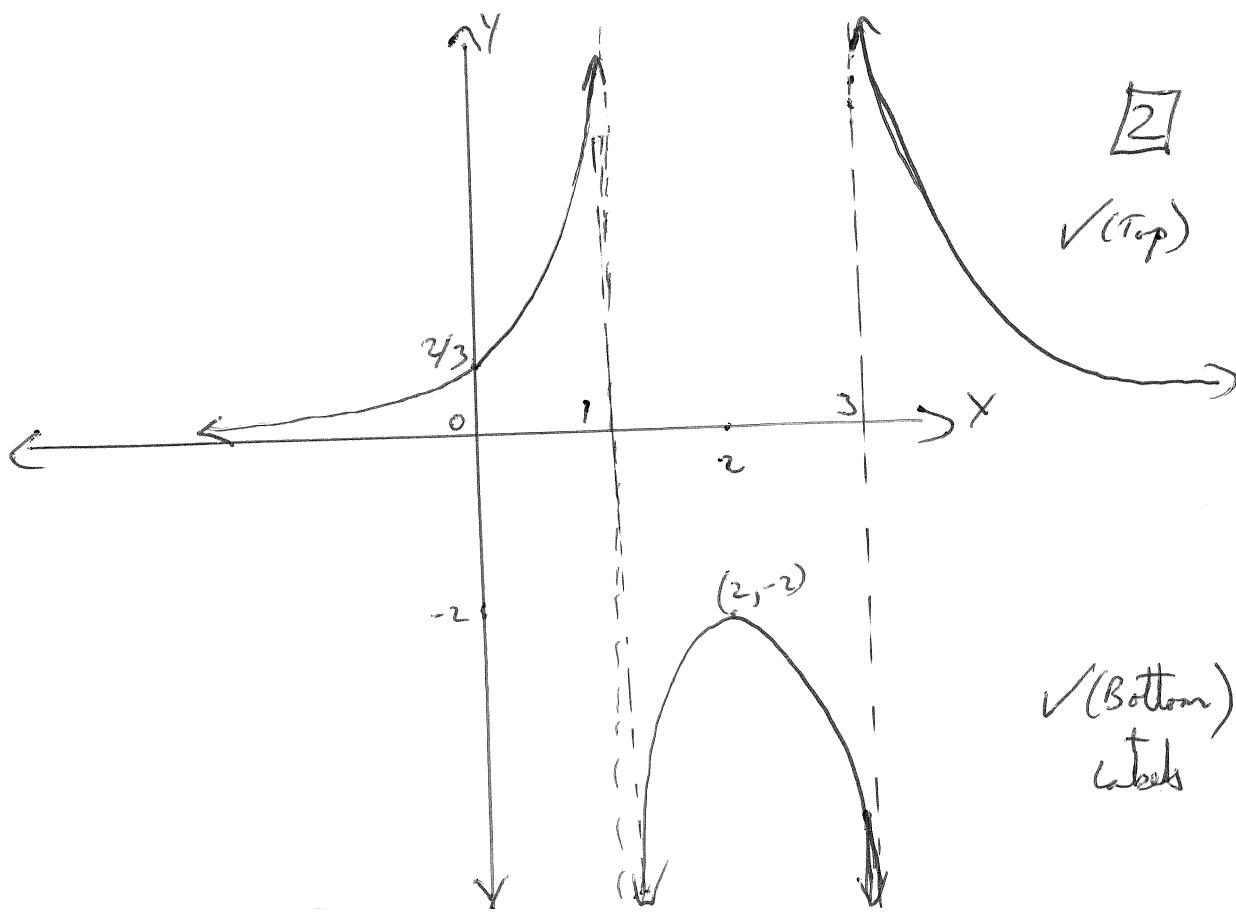
$\therefore$ Coordinates $(2, -2)$

$\therefore$ Max T.P.

Test:

✓

x	1.9	2	2.1
$\frac{dy}{dx}$	+	0	-
	/	-	\



[2]

✓ (Top)

✓ (Bottom)
Labels